File permissions in Linux

Project description

As a security professional, I examined and corrected file and directory permissions in a research team's Linux environment. I ensured that only authorized users had the appropriate access while following the principle of least privilege. This involved checking current permissions, identifying issues (such as overly permissive files and hidden files), and modifying them using Linux commands.

Check file and directory details

```
researcher2@04a51e2932fa:~$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 23:37 .
drwxr-xr-x 1 root        root          4096 Jul 17 22:45 ..
-rw------- 1 researcher2 research_team    6 Jul 17 23:37 .bash_history
-rw-r--r-- 1 researcher2 research_team  220 Mar 27  2022 .bash_logout
-rw-r--r-- 1 researcher2 research_team 3574 Jul 17 22:45 .bashrc
-rw-r--r-- 1 researcher2 research_team 3574 Jul 17 22:45 .profile
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 22:45 projects
researcher2@04a51e2932fa:~$ cd projects
researcher2@04a51e2932fa:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 22:45 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 23:37 ..
-rw--w---- 1 researcher2 research_team   46 Jul 17 22:45 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul 17 22:45 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jul 17 22:45 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul 17 22:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 22:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 22:45 project_t.txt
```

I used the command ***ls -la*** to view detailed file and directory information, including permissions, ownership, and hidden files. And ***cd*** to navigate between folders and directories.

The permissions string description

The 10-character permission string represents:

- Position 1: File type (- = file, d = directory)
- Positions 2-4: Owner permissions (r = read, w = write, x = execute)
- Positions 5-7: Group permissions

- Positions 8-10: Others permissions

Example: ***-rw-r--r--*** means the owner can read and write, while the group and others can only read.

Change file permissions

```
researcher2@04a51e2932fa:~/projects$ chmod o-w project_k.txt
researcher2@04a51e2932fa:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 22:45 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 23:37 ..
-rw--w---- 1 researcher2 research_team   46 Jul 17 22:45 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul 17 22:45 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 22:45 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul 17 22:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 22:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 22:45 project_t.txt
```

chmod o-w project_k.txt; this command removed write permission for others on project_k.txt.

Change file permissions on a hidden file

```
researcher2@79d4842049f6:~/projects$ chmod u-w,g-w,g-r .project_x.txt
researcher2@79d4842049f6:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 23:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 18 00:44 ..
-r-------- 1 researcher2 research_team   46 Jul 17 23:37 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul 17 23:37 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul 17 23:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_t.txt
```

chmod u-w,g-w,o-r .project_x.txt; this removed write permission from owner and group, and read permission from others on the hidden file .project_x.txt.

Change directory permissions

```
researcher2@79d4842049f6:~/projects$ chmod g-x drafts
researcher2@79d4842049f6:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 17 23:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 18 00:44 ..
-r-------- 1 researcher2 research_team   46 Jul 17 23:37 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jul 17 23:37 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul 17 23:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul 17 23:37 project_t.txt
```

chmod g-x drafts; This removed execute (traverse) permission from the group on the drafts directory to restrict access.

Summary

In this activity, I successfully managed Linux file and directory permissions to enhance system security. I practiced using *ls -la* to inspect permissions and *chmod* to modify them. These skills are essential for maintaining proper access control, protecting sensitive research data, and preventing unauthorized access in a Linux environment.